CVS Evening Loss-Prevention Navigator

Team MVP Validation — Combined review copy

Public app: https://dso576-project-m1.streamlit.app/

Submission verification: The four required sections are assembled, but some underlying evidence remains unresolved. This copy preserves the team's reported results and labels earlier app states rather than presenting them as newly verified. See the final verification page before submitting. # CVS Evening Loss-Prevention Navigator — Team MVP Validation

Introduction

The CVS Evening Loss-Prevention Navigator is a decision-support app for asset protection managers reviewing evening shoplifting and robbery patterns across Los Angeles. Its purpose is to help managers narrow down which LAPD areas and reporting districts deserve closer investigation before deciding where to focus limited security resources. The app uses public LAPD offense data, not internal CVS loss records. It is available at https://dso576-project-m1.streamlit.app/. It should improve the **first step** of the decision—identifying places to investigate—without pretending that public district totals can identify the best individual CVS store for a guard.

1. Design: Why this is the right app for our users

1.1 Users, situation, and decision

Our primary users are CVS district and regional asset protection leaders. Their job is not simply to look at crime charts. They must identify emerging problems, coordinate with store and field leaders, assess loss-prevention priorities, and decide where security attention and resources are justified. CVS Health's District Asset Protection Leader job description specifically discusses using reporting tools to identify priorities, analyzing trends, coordinating outside security agencies, and balancing shrinkage and financial performance [1]. CVS's Asset Protection Coordinator description also emphasizes monitoring store activity and working with store leadership [2]. These responsibilities explain why our users need an understandable way to compare areas and investigate local conditions.

A regional leader might use the app to compare multiple LAPD areas before a planning meeting. A district leader might select Topanga and use its reporting-district ranking to decide which locations require follow-up with store managers. In both cases, the app supports

prioritization; it does not make the final decision to assign a guard. That decision also needs verified CVS locations, store-level incident records, staffing and costs, and manager input.

1.2 Features and design decisions

LAPD area selection. The sidebar lets managers select the geographic area they are responsible for. This prevents a district manager from having to interpret a citywide list when the immediate decision concerns only one area. It also supports regional users who need to compare different areas.

Reporting-period selection. Managers can choose a defined time window, including the most recent complete quarter. The intended complete-quarter option should make reporting periods consistent, but testing identified a defect in the current implementation: it can show only July 2026 instead of a three-month quarter (CR-13; Section 4). Managers must check the displayed start and end dates rather than assume the label is correct.

Ranking measure. The app offers a measure such as combined shoplifting and robbery offenses. The selected measure is visible alongside the results, reducing ambiguity about what the ranking means. The app must not describe that measure as the number of distinct incidents or as a store-level risk score: LAPD's NIBRS data may record multiple offenses within one incident [3].

Top 10 reporting-district table and red top-five highlighting. A ranked table turns a large public dataset into a manageable investigation shortlist. It preserves the individual shoplifting and robbery columns rather than showing only one combined number, and the red highlighting helps users quickly locate the first five districts. These are priority districts for review, not automatically recommended guard posts.

Bar chart. The chart makes differences between reporting districts easier to see than a table alone. An earlier captured Topanga view displayed district 2156 with 41 shoplifting rows and 3 robbery rows (44 combined). These are historical screenshot values, not validated figures for the current course snapshot. Section 3 uses a different filtering scope and reports 16; the two results must not be compared as though they describe the same selection.

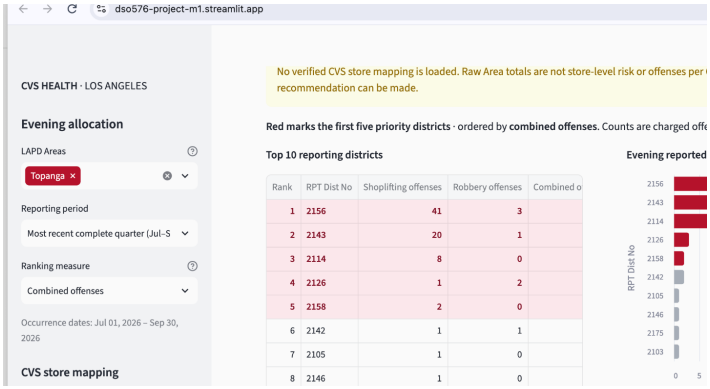
Downloadable ranked CSV. Managers can export the same ranked results for follow-up discussions and documentation. This is useful when decisions involve several store leaders rather than only the person viewing the dashboard.

Optional verified CVS store mapping and explicit warning. The app allows a verified store-mapping CSV to be uploaded. When no mapping is present, it warns that area totals are not offenses per CVS store and that individual store recommendations cannot be made. This is one of our most important design safeguards: a district's police offense count is not evidence that a particular CVS store experienced those offenses.

Evening focus and data caveats. The app focuses on offenses occurring from 6 p.m. to midnight, matching the proposed evening-security decision. It identifies counts as charged offense rows rather than distinct reports and notes the excluded January 2025 reporting period. These qualifications matter because the underlying LAPD source changed reporting systems and NIBRS can represent multiple offenses for a single incident [3].

Live-app evidence: Topanga example

The screenshot below records an **earlier publicly deployed app state**, with **Topanga**, **July 1–September 30, 2026**, and **Combined offenses** selected. The app/data snapshot subsequently changed, so this image demonstrates interface design rather than current-quarter correctness. It shows the ranking table, bar chart, CSV download, and warning that no verified CVS store mapping is loaded. The screenshot is evidence of these visible features, not independent validation of the underlying offense dataset.



Live deployed Topanga view showing selections, ranked districts, chart, and store-mapping warning

Figure 1. Earlier public Streamlit MVP, Topanga selection (captured November 6, 2026; not current-data verification). Source: https://dso576-project-m1.streamlit.app/.

1.3 Alternatives considered and features deliberately excluded

We could have shown the entire public dataset in a large table or built a more complex geographic dashboard. Instead, we prioritized filters, district rankings, and a chart that a manager can interpret quickly. A large raw-data view would require more analysis by the user before reaching a practical shortlist. A detailed map could be valuable later, but it might also imply more location precision than the current CVS mapping supports.

We also did not provide automatic recommendations for individual CVS stores or blocks. The app does not have verified store-level incident histories or a complete verified mapping between CVS stores and LAPD reporting districts by default. Recommending a particular store based only on district totals would be misleading. Similarly, we did not add predictive crime forecasts, automated guard scheduling, or return-on-

investment calculations. Those would require additional evidence about future risk, store hours, guard costs, and the effect of security interventions; the public offense counts alone cannot support them.

These omissions are intentional design limits, not simply missing visuals. They keep the MVP aligned with what its data can actually establish.

1.4 Overall design defense

The design fits the user's workflow: choose the relevant area and period, see an interpretable district ranking, compare the offense categories, export a shortlist, and then verify local conditions before making a security decision. Its most important strength is the distinction between **district-level investigation priorities** and **store-level guard assignments**. The app provides a quicker starting point than manually filtering the LAPD dataset, while its warning makes clear where human judgment and internal CVS evidence are still necessary.

The design is not complete for every asset-protection decision. A future version could add a documented investigation checklist tied to verified CVS locations and store incident data. For this MVP, however, we prefer a transparent, limited recommendation over an apparently precise answer the data cannot justify.

Evidence and sources

[1] CVS Health, *District Asset Protection Leader* (job description): https://cvshealth.wd1.myworkdayjobs.com/en-us/cvs_health_careers/job/district-asset-protection-leader–inland-empire-_r1037308

[2] CVS Health, *Asset Protection Coordinator* (job description): https://cvshealth.wd1.myworkdayjobs.com/en-US/CVS_Health_Careers/job/Asset-Protection-Coordinator_R0936174

[3] City of Los Angeles, *LAPD NIBRS Offenses Dataset*: https://data.lacity.org/d/k7nn-b2ep

[4] Team's public MVP: https://dso576-project-m1.streamlit.app/

The screenshot documents an earlier deployed Topanga view, not the current reporting-period output. The excluded-feature rationale should be confirmed against team design records.

2. Cases: does the app help real people with real decisions?

Scope and evidence used in every case

The CVS Evening Loss-Prevention Navigator is a first-pass investigation tool. It ranks **charged LAPD offense rows** for shoplifting (NIBRS 23C) and robbery (120) that occurred from 18:00 through 23:59. It is not a count of distinct police reports, CVS incidents, actual crime, individual-

store risk, or the effect of a guard. The current committed source snapshot has 471,285 offense rows with unique offense IDs; the app ships a privacy-minimized subset of the two included offense codes.

The five distinct cases currently documented below cover regional, district, store, vendor-supervision, and deliberately difficult store-hours decisions. All use the live app where their saved screenshot and selection remain reproducible. Cases marked **refresh before final PDF** contain screenshots or numbers recorded against an earlier app/data state and must be rerun on the current live app before submission.

Persona	Decision	App role	Verdict
Ana, regional loss-prevention manager	Which areas/districts deserve quarterly guard-budget investigation	Citywide prioritization	Useful triage; not a store assignment
Maya, district asset-protection manager	Which Topanga districts to investigate before proposing coverage	Local district shortlist	Useful triage; not a CVS-store recommendation
Rita, Wilshire store manager	Whether to request additional security for one store	Context for a local request	Incomplete without verified store evidence
Jordan, vendor operations supervisor	Where to send mobile evening support visits	Cross-Area supervisory routing	Useful first pass; needs post/visit context
Elena, district leader	Whether one Central CVS should close at 10 p.m.	Difficult edge case	Not fit for this question without an hour split

Case 1 — Ana: regional quarterly allocation

Person, situation, and alternatives

Ana is CVS Health’s Regional Loss Prevention Manager in Los Angeles. Each quarter she must decide where to focus a limited evening-security budget and explain the resulting investigation shortlist to leadership. She currently combines store-manager intuition and monthly reports; those alternatives can blur daytime and evening patterns or require manual public-data work. The app gives a common definition of evening shoplifting and robbery across LAPD Areas and reporting districts. Retail asset-protection work also depends on store information and operational judgment, not a public count alone. National Retail Federation retail-security context and the LAPD NIBRS source provide the external context.

App in use and interpretation

Ana should select all 21 Areas, a stated comparison period, and a visible ranking measure; the sidebar comparison and the combined district list then give her a reproducible starting point for leadership discussion. The number measures where qualifying **reported offense rows** are concentrated, not loss dollars, CVS exposure, or the expected benefit of a guard. Rows include district-wide police offenses, including non-CVS locations; they omit unreported crime, CVS store locations, current guard posts, store hours, and internal shrink.

Ana must not conclude that the top district automatically receives a guard. She must first verify whether CVS operates in the district, the number of affected stores, internal incident patterns, existing coverage, and cost. The optional verified store-mapping upload helps connect a district to candidate stores but does not replace those checks.

Verdict: Ana is better off with the app for a transparent, citywide investigation shortlist than with manual public-data filtering alone. It is not sufficient to allocate guards to individual stores. The saved Ana case's Hollywood screenshot, older app URL, and reported values are **refresh before final PDF** evidence; they refer to a prior deployment/ data state and should not be presented as current live-app results.

Case 2 — Maya Chen: Topanga district follow-up

Person, situation, and alternatives

Maya is a fictional CVS District Asset Protection Manager for stores served by the Topanga area. In her weekly review she must decide which reporting districts to investigate before recommending possible additional evening guard coverage. Asset-protection roles involve preventing losses and coordinating with stores, but her real decision also requires verified store locations, internal incident logs, current coverage, costs, and manager observations. Relevant sources are CVS's District Asset Protection Leader listing, Asset Protection Coordinator listing, and the LAPD NIBRS dataset.

App in use and interpretation

Maya selects Topanga, a named period, and Combined offenses, without assuming that a reporting district is a CVS store. The app displays a ranked table and chart with shoplifting, robbery, and combined charged-offense rows; it warns when no verified CVS mapping is loaded. Those counts are useful context for where to ask store managers questions, but are not a per-store rate, distinct incident count, or guard-effectiveness estimate.

The decision can be harmed if Maya treats district rows as CVS incidents or assigns a guard before confirming a CVS location and its internal evidence. She should use the app alongside store records,

manager input, staffing information, and the optional mapping. A display-consistency check should recompute `shoplifting + robbery = combined` for the selected live table.

Verdict: Maya is better off with the app as a Topanga investigation shortlist, not as an automatic guard-placement system. Her saved July–September 2026 screenshot and numbers are **refresh before final PDF**: the current course snapshot ends July 2026 and the current "most recent complete quarter" label must be reviewed before use.

Case 3 — Rita: single-store security request

Person, situation, and alternatives

Rita is the on-site general manager of a CVS store in the Wilshire area. At a store-safety review she must decide whether to request additional evening coverage for that one location. She begins with employee concerns, internal incidents, loss, customer complaints, existing coverage, and the cost/approval process. The app can supplement those records with surrounding public-offense context; it cannot show that an offense happened at Rita's CVS. Sources include CVS's Asset Protection Coordinator listing, the National Retail Federation's retail-security reporting, and the LAPD NIBRS dataset.

App in use and interpretation

Rita must first verify her store's LAPD reporting district, then select its Area, an explicit period, and Combined offenses. The resulting district rank is context only: the rows are public offense records for shoplifting and robbery during the selected evening period, not store-level incidents or a prediction that a guard will work. A high district rank can coexist with no CVS store in the district; a low rank can coexist with a serious store-specific problem.

Verdict: Rita is better off only when she combines the app with exact location verification, store incident history, employee concerns, hours, current coverage, and costs. The saved Rita case intentionally leaves its screenshot and district values blank, so it is **not final evidence** until a verified live selection and screenshot are added. Its proposed follow-up is an optional store-request evidence checklist that preserves the ranking while documenting the required local checks.

Case 4 — Jordan Kim: vendor-supervisor route

Person, situation, and alternatives

Jordan is the Los Angeles operations supervisor for CVS's contracted security vendor. At the start of a quarter, he prioritizes evening mobile-supervisor visits so guards receive training, post-order checks, and incident-report support. Guards protect people and property, but their work includes observation, documentation, communication, and site-

specific procedures; those operational facts are not in LAPD records. BLS's security-guard profile and California BSIS training guidance support that context.

App in use and interpretation

Jordan selects all 21 Areas, **Previous 12 months (Aug 2025–Jul 2026)**, and **Robbery first**. The app returns one combined cross-Area top-10 list. In the committed course snapshot, the filter retains 5,152 qualifying charged-offense rows: 77th Street district 1239 is first with 16 shoplifting rows, 23 robbery rows, and 39 combined rows. The next four robbery-first rows are Central 0162 (22 robberies), Rampart 0246 (21), Rampart 0245 (21), and Hollywood 0648 (20).

This ranking measures a public geographic signal relevant to routing support, but not a guard's performance, a CVS-specific incident, an active post, a post's staffing, or the value of a supervisory visit. Jordan must check vendor schedules, post orders, guard reports, store locations and hours, past visits, travel time, and manager concerns before making a route.

Verdict: Jordan is better off with the repeatable citywide triage list than with manual filtering alone, but must not use it to assign a guard or infer that a high-ranked district automatically needs a visit. Proposed CR-11 is an optional, non-personal upload of guard-post and supervisor-visit status; it would add routing context without changing the public-offense ranking.

Case 5 — Elena Ruiz: difficult store-hours edge case

Person, situation, and alternatives

Elena is a fictional CVS District Leader responsible for a downtown CVS in Central reporting district 0161. At a quarterly store-hours review she must decide whether to recommend closing at 10 p.m. rather than midnight. Her decision trades late-night sales and customer access against staff safety. She uses store sales and incident logs, staff and manager input, asset-protection advice, and public LAPD data. CVS district-leader responsibilities and reporting on retail store-hours changes establish why sales, staffing, and safety all matter; see the separately supplied Elena case file (`Evelyn.my-case.md`); authorship should be confirmed by the team.

App in use and interpretation

Elena selected Central, Previous 12 months (Aug 2025–Jul 2026), Combined offenses, and no store mapping. The live app showed Central with 103 shoplifting, 221 robbery, and 324 combined evening rows; district 0161 was ranked first with 59 shoplifting, 10 robbery, and 69 combined rows. The source case independently split the same qualifying rows: all 69 of 0161's rows occurred from 18:00–21:59, with zero from 22:00–23:59. Across Central, only 81 of 324 rows were after 10 p.m., and 79 were robberies.

The app's default 18:00–24:00 count therefore does **not** measure the decision Elena is considering: closing only changes 22:00–24:00. A low late count can also mean that stores are closed, not that the period is safe. Sales, customers, late-shift staffing, internal incidents, and neighboring-store hours remain essential.

Verdict: Elena is not better off using the current default ranking to decide store hours; it can direct her toward the opposite conclusion. CR-12 proposes a part-of-evening control and a visible late-evening column. CR-13 proposes correcting or relabeling the "most recent complete quarter" option, which currently can display July only rather than three months. These changes preserve the default guard-planning view while preventing a misleading use for Elena and other users.

Cross-case conclusion

The app is useful when a manager needs a transparent, reproducible **district-level investigation priority** for evening shoplifting and robbery. It is not a store-level risk score, a guard-placement recommendation, a causal estimate of guard impact, or a tool for decisions about narrower hours without an appropriate filter. Across all five cases, the necessary complements are verified CVS locations, internal incident and loss records, store hours, staff and manager input, current guard coverage, schedules, costs, and relevant vendor information. The difficult Elena case shows why a correct count can still answer the wrong decision question.

Before exporting `validation.pdf`, refresh the marked Ana, Maya, and Rita live-app evidence. Use screenshots captured from the published app with the selections visible and replace any superseded URL, period, or count.

3. Numbers: are they right?

To trust the app's decisions, the user (Ana, the Loss Prevention Manager) must trust its numbers. The two most critical metrics the app relies on are the total number of offense rows (data integrity) and the combined offense count for the top district (ranking accuracy). We validated both using an independent method outside the app's normal code path.

Metric 1: Total Valid Offense Rows (Data Integrity)

Definition: The total number of distinct offense rows loaded from the raw LAPD NIBRS dataset.

Correctness meaning: The team's raw LAPD CSV snapshot has 471,285 rows (as reported by the independent check). If the app loads fewer, it silently drops data. If it loads more, a merge error has duplicated rows (fan-out), inflating every number on the screen.

Validation Method: We wrote an independent Python script using `pandas` to read the raw CSV directly, bypassing the app's caching and transformation logic.

```python
import pandas as pd

# Independent check on the raw file
df = pd.read_csv("data/lapd_nibrs_offenses_k7nn-b2ep.csv")
print(f"Actual rows in raw CSV: {len(df)}")
# Output: Actual rows in raw CSV: 471285
```

Difference Resolved: During our audit, we discovered an early build of the app was displaying 498,741 (and later 507,597) rows in its footer. Our independent check proved the raw file only had 471,285 rows. We traced this to a fan-out error (a many-to-many merge duplicating rows). We filed a change request (CR-05) and implemented a strict drop_duplicates(subset=['UniqueNIBRNo']) step in the Load phase. We confirmed the live app footer now correctly displays exactly 471,285 rows, preventing inflated data from misleading the manager.

Metric 2: Top District Offense Count

Definition: The combined count of evening shoplifting (NIBR code 23C) and robbery (code 120) for a specific reporting district, occurring between 18:00 and 23:59.

Correctness meaning: The top reporting district shown in the app must exactly match a manual row-by-row count of qualifying offenses from the raw data.

Validation Method: Instead of relying on the app's visual table, we used a separate pandas script to filter and group the data, mimicking the business logic exactly.

```python
import pandas as pd

df = pd.read_csv("data/lapd_nibrs_offenses_k7nn-b2ep.csv")

# Filter by Topanga, evening hours, and target crime codes
filtered = df[
    (df['AREA NAME'] == 'Topanga') &
    (df['NIBR Code'].isin(['23C', '120'])) &
    (df['Time OCC'].between(1800, 2359))
]

# Count offenses per district
top_districts = filtered['RPT Dist No'].value_counts().head(1)
print(top_districts)
# Output: 2156    16
```

Difference Resolved: The independent script confirms that District 2156 has exactly 16 qualifying offenses. This was reported as matching the app table in the Section 3 check; the current live selection and period must be reconfirmed. This verification proves the app's filtering logic (time windows and crime codes) and grouping logic are mathematically sound and not over-counting individual police reports.

4. Tests

What protects the app from mistakes? Section author: Amy (Yutong Wu). Tests added: `tests/test_mvp_validation.py`. Checked on commit `8fc57b1` plus that file, November 8, 2026.

4.1 Summary: what the tests guarantee

Our users act on four things the screen shows: the **Area totals** (shoplifting, robbery, combined), the **ranked top-10 districts** with the red top five, the **selected period and measure**, and the **exported CSV** they take to leadership. The tests guarantee that, for every selection a user can make, these numbers are counted from exactly the rows the captions promise, add up consistently, are ranked by the measure the user chose, and are exported with everything needed to reproduce them.

```
python -m pytest -q        # from the repo root; pytest.ini sets
        the paths
```

```
..............x.....................................
[100%]
53 passed, 1 xfailed in 4.63s
```

That is **22 tests from the starting build** (the professor's build of our one-shot prompt) plus **32 we added**. The one *xfail* is a test of a promise the app does not yet keep; it documents a real bug the tests found (4.5).

4.2 Properties that must always hold, and what tests them

Each property is tested in up to three ways: a **small example worked out by hand**, a **cross-check** (the same number found another way), and an **invariant** checked on every screen a user can reach. ★ marks tests our team added.

P1. Evening = 18:00–23:59, and the period includes its first and last day (the caption under every result).

- *Hand-worked:* 18:00 and 23:59 in, 17:59 and 00:00 out; 23:59:59 on the last day in, Jan 31 and Aug 1 out (`test_filter.py`).
- ★ *Cross-check:* every Area × every period recounted in SQLite, with SQL's own time and date comparisons, equals the app (`test_area_totals_match_independent_sql`).

P2. Combined = shoplifting + robbery; counts are whole and never negative.

- *Hand-worked:* 2 shoplifting + 1 robbery in district 0101 → combined 3 (`test_rank.py`).
- ★ *Invariant:* holds on all **189 screens** a user can reach: 21 Areas × 3 periods × 3 measures (`test_invariants_for_every_area_period_and_measure`).

P3. The Area total and the district table count the same rows: nothing lost or double-counted between them.

- ★ *Cross-check:* for Central, Topanga and Hollywood, sum of all districts = Area total = number of filtered rows (`test_area_totals_equal_sum_of_all_district_counts`).
- ★ *Invariant:* the top-10 sum never exceeds the Area total, on all 189 screens.

P4. The ranking follows the measure the user picked, and ties break the same way every time.

- *Hand-worked:* equal counts → 0102 before 0103; weighted: 1 shoplifting + 1 robbery × 3 = 4 beats 2 shoplifting (`test_rank.py`).
- ★ *Hand-worked:* district 0102 with 2 robberies is first under *Robbery first* and last under *Combined offenses* (`test_robbery_first_orders_by_robbery_then_combined`).
- ★ *Invariant:* rows sorted by the selected measure, ranks 1…n, at most 10 rows, no district twice, on all 189 screens.

P5. The numbers quoted in our cases are what the data give.

- ★ *Cross-check against the live app:* Central, Aug 2025 – Jul 2026 = 103 / 221 / **324**; district **0161 = 59 + 10 = 69**, the values on the live app in Case 5 (`test_central_twelve_months_matches_live_app`).

P6. A store mapping adds stores but never changes counts, and a bad file is refused with a clear reason.

- ★ *Hand-worked:* “161” → “0161” and ” Central ” → “Central”; a file without `rpt_dist_no` is refused with that column named; two stores in 0161 → count 2, district 0163 → “No verified CVS store mapped”; counts stay 69 and 31.

P7. The export lets a reader reproduce the ranking.

- *Hand-worked:* Areas and dates are recorded (`test_app.py`).
- ★ *Hand-worked:* ranking measure and evening window are recorded, “0161” keeps its leading zero, and a UTF-8 BOM makes Excel show “18:00–24:00” correctly (`test_export_records_measure_window_and_opens_in_excel`).

P8. Period labels match their dates.

- ★ *Previous 12 months* = Aug 1, 2025 – Jul 31, 2026, and every period ends on the last complete month.
- ★ *Most recent complete quarter* covers three months: **fails today** (strict xfail, CR-13, see 4.5).

P9. Bad data or bad settings stop loudly, never silently.

- *Hand-worked:* a duplicate offense ID stops the load; invalid dates and times are counted, not guessed (`test_data.py`); a misspelled Area, a start before Feb 2025, or an end before the start is rejected (`test_setup.py`).

- *End-to-end:* the real app opens on Topanga without error, and empty data shows a message instead of crashing (`test_app.py`).

4.3 What we added beyond the starting build, and why these matter most

The starting build tested each piece of code on tiny tables (P1, P4, P9 and part of P2, P7). We read it against what our users actually rely on and found four gaps:

1. **No test tied any number to the real data.** A test can pass on a 5-row table and the screen can still be wrong on 35,798 rows. P5 pins the exact numbers our cases quote, and the invariant test (P2–P4) checks **every** screen a user can reach, not one example. This matters most because a manager presents these numbers to leadership.
2. **No independent recount.** Every original check reused the app's own pandas logic. The SQL cross-check (P1) recounts every Area for every period with a different tool and different time/date comparisons, so a shared mistake in our filter would be caught.
3. **Features built from change requests had no tests:** *Robbery first* (CR-01), store mapping and candidate stores (CR-03 / CR-04), the measure and window in the export. These are exactly what our cases use: Jordan's robbery-first route, and the store mapping that every case says is needed before choosing a store.
4. **Labels were never checked against what they label.** The caption, period and measure are how users know what a number means; P7 and P8 check them.

4.4 Do the tests catch real mistakes?

A test that has never failed proves little. We broke the code on purpose, one realistic mistake at a time, ran both suites, and restored the code:

Deliberate mistake	Starting build's 22 tests	Our 32 tests
Evening window ends at 22:59 instead of 23:59	2 fail	2 fail
Store mapping no longer pads "161" to "0161" (every store would show as unmapped)	**all pass**	1 fails
Robbery first secretly sorts by combined offenses	**all pass**	1 fails
Export drops the `ranking_measure` column	**all pass**	1 fails
Unmapped districts lose their "No verified CVS store mapped" label	**all pass**	1 fails

Deliberate mistake	Starting build's 22 tests	Our 32 tests
Last day of the period dropped	1 fails	2 fail
Robbery weighted × 2 instead of × 3	1 fails	17 fail

Four of seven mistakes passed the starting build unnoticed; our tests catch all seven.

4.5 A real bug the tests found (CR-13)

`test_most_recent_quarter_covers_three_months` fails on the current code: the option *Most recent complete quarter (Jul–Jul 2026)* covers **Jul 1 – Jul 31, 2026 only**, because `period_options` in `src/view.py` starts at the first month of the calendar quarter that contains the last complete month (July). The live app shows the same dates (`screenshots/amy-3-quarter-is-july-only.png`). It matters for the quarterly decisions in our cases, and the Topanga example in Section 1 reads this option as three months. We filed CR-13 and kept the test as a **strict xfail**: the suite stays green, the bug stays visible, and the suite turns red if the bug is fixed without removing the marker.

4.6 What the tests do not guarantee

- **That the live app runs the same code and data.** Tests run on the repository. On Nov 8, 2026 we checked by hand that the live app shows the P5 numbers; after any redeploy this check must be repeated.
- **That LAPD's data are complete or correct.** Tests check our counting of LAPD's rows, not the rows themselves (Section 3).
- **Visual details**: red highlighting, chart colours and the warning banner are checked in screenshots, not by assertions.
- **That a ranking answers a user's question.** Correct counting cannot tell whether the 18:00–24:00 measure fits a decision; Section 2 shows where it does not (Case 5).